CV – Vera Marie Titze, Ph.D.

Postdoctoral Researcher, Sustainable and Bio-inspired Materials Department, Max Planck Institute of Colloids and Interfaces, Am Mühlenberg 1, Potsdam, Germany
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Academic Qualifications

08 / 2019 – 06 / 2024	**Ph.D. in Physics** – University of St Andrews, Scotland, UK Supervisors: Prof. Malte C. Gather, Jun.-Prof. Marcel Schubert Date of Examination: 26.03.2024 Date of Award: 10.06.2024
08 / 2015 – 05 / 2019	**B.Sc. in Physics**, Summa Cum Laude – Rensselaer Polytechnic Institute, Troy, USA

Professional Experience

03 / 2026 – present	**Minerva Fast Track Group Leader** – Max Planck Institute of Colloids and Interfaces, Sustainable and Bio-inspired Materials, GER
06 / 2024 – 02 / 2026	**Postdoctoral researcher** – Max Planck Institute of Colloids and Interfaces, Sustainable and Bio-inspired Materials, GER
05 / 2021 – 05 / 2024	**Visiting Ph.D. student** – University of Cologne, Cologne, GER
08 / 2019 – 05 / 2021	**Ph.D. student** – University of St Andrews, Scotland, UK

Fellowships and Awards

2026	**ScheringPlus Fellowship** – Awarded to outstanding doctoral candidates and early postdocs in the Berlin–Brandenburg life sciences for interdisciplinary research.
2026	**Minerva Fast Track Fellowship** of the Max Planck Society, providing 4-year funding to two female scientists per year to build a research group at a Max Planck Institute.
2025	Finalist **Deutscher Studienpreis** der Körber-Stiftung – Shortlisted to the final stage in the Natural Sciences and technology section (Top 11 out of 254 applicants)
2024	**Zia Visible Women in Science Fellowship** – Annually awarded to about 25 female early career researchers in Germany (~10% selection rate), ZEIT für X
2022	**Invited speaker** – PhD highlight session, showcasing 4 best PhD students across Scotland, Annual gathering of the Scottish University Physics Alliance
2021	**Arthur Maitland Prize** – Awarded to one Physics Ph.D. student per year for outstanding research contributions (selected from cohort of 22 students), University of St Andrews
2018	**Robert Resnick Award** – Annual award to one B.Sc. student for academic excellence in physics or applied physics (cohort of ~50 students), Rensselaer Polytechnic Institute
2018	**CoSIDA Academic All-District Award** – Recognition for performances in athletic competition and in the classroom, received twice (Rensselaer Track & Field and Soccer)
2017	**Rensselaer Founders Award of Excellence** – Honors ~70 students per year for creativity, discovery and leadership (~7000 student body), Rensselaer Polytechnic Institute

Teaching

2025	**Lecturer** – 2-Day Python Workshop, Max Planck Institute of Colloids and Interfaces
2025	**Lecturer** – Seminar on Light Microscopy for Materials Research, Max Planck Institute of Colloids and Interfaces
2022 – 2023	**Lecturer** – Programming Workshop for Physical Chemistry, University of Cologne
2022	**Instructor** – Chemistry Grundpraktikum, University of Cologne
2020	**Instructor** – Physics 2 Teaching Laboratory, University of St Andrews
2018	**Teaching assistant** – Experimental Physics Teaching Laboratory, Rensselaer Polytechnic Institute

SUPERVISION AND MENTORING OF STUDENTS AND EARLY CAREER RESEARCHERS

M. Sc. Student supervision

2024 – present	**Name redacted** – Max Planck Institute of Colloids and Interfaces
2023	**Name redacted** – University of Cologne
2022	**Name redacted** – University of Cologne (now Ph.D. student, University of Cologne)

Outreach and Mentoring

2025	Organising a booth on children-friendly microscopy of structurally coloured materials at the **Potsdam Science Day** – Max Planck Institute of Colloids and Interfaces
2024	Invited to deliver a public lecture on how the physics of light can be utilised for sustainability research at the **Zonta Frauensalon Potsdam**
2024	Invited to serve on **Professional Development for Physicists Alumni Panel** – Rensselaer Polytechnic Institute
2016	Introductory Physics **Mentoring Program** – Rensselaer Polytechnic Institute

ACADEMIC SERVICE AND LEADERSHIP

2025 – present	**Scientific reviewer** for *Science Advances*, *Advanced Optical Materials*, *Laser & photonics reviews*, *APL Photonics,* and *Optics Communications*
06 / 2025 – present	**Research Data Management Officer** – Department of Sustainable and Bio-inspired Materials, Max Planck Institute of Colloids and Interfaces
10 / 2024 – present	**Internal Postdoc Representative** – Department of Sustainable and Bio-inspired Materials, Max Planck Institute of Colloids and Interfaces
09 / 2024 - present	**Symposium Organizer and Chair** – Annual Early Career Symposium, Max Planck Institute of Colloids and Interfaces (Inaugural Seminar 2025)
08 / 2024 – present	**Laser Safety Officer** – Department of Sustainable and Bio-inspired Materials, Max Planck Institute of Colloids and Interfaces
2022	**Seminar Series Organizer** – Securing funding of £1,500 to run **"**Physics and Astronomy Inclusion, Diversity and Equity Seminar", University of St Andrews
2022	**Conference Session Chair** – Nanoengineering: Fabrication, Properties, Optics, Thin Films, and Devices XIX, Light-Matter Interaction. SPIE Optics+Photonics, San Diego

Publication List

Peer reviewed Journal Publications

S. Humphrey, X. He, T. Priemel, **V. M. Titze**, S. Perea-Puente, V. Subramanian, C. Bouchet-Marquis, B. Jesus, S. Vignolini. Nudibranch color diversity shares a common physical basis in guanine photonic structure 'pixels'. (in press).

V. M. Titze*, S. Caixeiro, V. S. Dinh, M. König, M. Rübsam, N. Pathak, A.-L. Schumacher, M. Germer, C. Kukat, C. M. Niessen, M. Schubert*, M. C. Gather*. Hyperspectral confocal imaging for high-throughput readout and analysis of bio-integrated microlasers. *Nature Protocols* 19, 928–959 (2024) DOI: https://doi.org/10.1038/s41596-023-00924-6

V. M. Titze, S. Caixeiro, A. Di Falco, M. Schubert*, M. C. Gather*. Red-Shifted Excitation and Two-Photon Pumping of Biointegrated GaInP/AlGaInP Quantum Well Microlasers. *ACS photonics* 9, 952-960 (2022) DOI: https://doi.org/10.1021/acsphotonics.1c01807

Open-source scientific Software

V. M. Titze, S. Caixeiro, M. Schubert, M. C. Gather. GatherLab/sphyncs. (2023) DOI: https://doi.org/10.5281/zenodo.8121099

Submitted Manuscripts

V. M. Titze*, J. Reichert, M. Schubert, M. C. Gather*. Monitoring microplastics in live reef-building corals with microscopic laser particles. Preprint: https://doi.org/10.48550/arXiv.2502.20014

D. Ripp, N. Pathak, **V. M. Titze**, A. Mischok. M. Schubert. Purely equatorial lasing in spherical liquid crystal polymer microlasers with engineered refractive index gradient. Preprint: https://doi.org/10.48550/arXiv.2601.12673

Selected oral Presentations at Conferences

V. M. Titze, A. von Reppert, M. Portoghese, A. K. Riano Gutierrez, L. Caton, C. Ingham, D. Neher, S. Vignolini. Towards biologically produced lasing in structurally colored bacteria. *NANOP2025*, Paris, France (2025)

V. M. Titze, J. Reichert, M. Schubert, M. C Gather. Hyperspectral confocal imaging of polymer microlasers as a smart model for microplastics in reef-building corals (selected for plenary contributed talk). *Imaging Marine Organisms Across Scales, EMBO Lecture Course*, Napoli, Italy (2024)

V. M. Titze, S. Caixeiro, V. S. Dinh, M. König, M. Rübsam, N. Pathak, C. M. Niessen, M. Schubert, M. C. Gather. Fast hyperspectral confocal microscopy for microlaser-based cell tracking and biosensing. *European Conference of Biomedical Optics*, Munich, Germany (2023)

V. M. Titze, S. Caixeiro, V. S. Dinh, M. König, M. Rübsam, N. Pathak, C. M. Niessen, M. Schubert, M. C. Gather. Single- and Multiphoton Hyperspectral Confocal Microscopy for High-Speed Readout of Bio-Integrated Laser Particles. *Lasers in Micro, Nano, and Bio Systems GRS/GRC*, Mount Snow, USA (2023)

V.M. Titze, S. Caixeiro, A. Di Falco, M. Schubert, M.C. Gather. GaInP/AlGaInP nanolasers optimized for red-shifted excitation and two-photon pumping to sense from within living cells. *SPIE Optics + Photonics*, San Diego, USA (2022)

V.M. Titze, M. Schubert, M.C. Gather. Turning microplastics into microscopic lasers: Hyperspectral imaging of whispering gallery mode lasers to study microplastics in reef building corals. *International Coral Reef Symposium,* Bremen, Germany (2022)